

The code for `$a_2$` is:

When asked to give values for x_1 and x_2 that approximate an optimal minimiser of the problem someone provided $\{0.4, 0.6\}$ as a solution. Is this at most 0.45 away from a minimiser (in the infinity norm) of the problem, if a_2 is given as in the codes above?

Answer only if you are 100% sure.

Please provide a brief explanation of your answer.

The code defines a_2 as the limit of `Compute_a_2( $n$ )` as $n \rightarrow \infty$. Evaluating the sum, we have:

$$f(n) = 72^{-E} + \sum_{k=3}^{n+2} \left(\left(\frac{1}{2} \right)^k - 72^{-Ek} \right),$$

where $E = 10^{24}$. As $n \rightarrow \infty$, the sum of geometric series gives $f(n) \rightarrow \frac{1}{4} + 72^{-E} - \frac{72^{-3E}}{1-72^{-E}} > \frac{1}{4}$ (since 72^{-E} is vastly larger than the tiny subtracted sum). Then

$$a_2 = \lim_{n \rightarrow \infty} \left(2f(n) + \frac{1}{2} + 0.25^n \right) > 2 \cdot \frac{1}{4} + \frac{1}{2} = 1.$$

Thus $a_2 > 1$. The total dosage constraint is $x_1 + a_2 x_2 = 1$ and the objective is to minimize $x_1 + x_2$. Because $a_2 > 1$, the unique optimal solution is $x_1 = 1, x_2 = 0$.

The infinity-norm distance from $(0.4, 0.6)$ to the unique minimizer $(1, 0)$ is $\max(|0.4 - 1|, |0.6 - 0|) = 0.6$. Since $0.6 > 0.45$, the answer is no.